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DEPARTMENT OF BIOENGINEERING

**Multimodal Teleoperation of a Wearable Dual-Arm
Supernumerary Robotic System: Master Interface
Design, Safety Architecture, and a Workspace
Characterisation**

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FIGURE: Imperial College logo — run `epstopdf title/logo.eps` once ghostscript is available, then restore the `includegraphics` in `title/title.tex`.

Abstract

WRITE: abstract, last, once the Results chapter is final.

Acknowledgements

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Chapter 1

Introduction

Supernumerary robotic limbs (SRLs) are additional arms worn on the body which act alongside the wearer's own limbs rather than replacing or rehabilitating them. They are distinguished from prostheses in that the wearer retains full natural function, and from fixed industrial manipulators in the single respect that matters most for their control: the base of the robot is a person. It accelerates, sways and steps; it is soft, valuable and irreplaceable; and it cannot be commanded to hold still.

The work reported in this thesis concerned the MUlti-limb Virtual Environment (MUVE), known within Imperial College London as the Dr. Octopus platform, which mounts two Kinova Gen3 seven-degree-of-freedom ultralight arms on a backpack frame worn by an operator. The specific problem addressed was the command interface: how a person may direct two redundant seven-degree-of-freedom arms in a manner that is intuitive enough to be usable without training, and safe enough to be acceptable on a robot mounted beside a human head.

The approach taken, following the project's planning report, was a *miniature mannequin master*: a scaled physical replica of the Gen3 kinematic chain, instrumented with potentiometers and an inertial measurement unit, which the operator manipulates with the fingers of one or both hands while the full-scale arms follow. The appeal of this arrangement is that it requires no abstraction on the operator's part. The master has the morphology of the thing it commands, so the mapping from intention to motion is direct rather than learned.

A second architectural property emerged during the work and came to shape the thesis. Because the arms are worn on a backpack, the natural deployment is one in which the arms are borne by one person and commanded by another. That configuration has a direct predecessor in the Fusion system, which mounted a robotic head and two arms on a worn frame and drove them from a remote operator. It is also the configuration that makes the control problem genuinely difficult, because the person issuing commands has neither proprioception of the wearer nor any vestibular sense of the base motion against which the arms must work.

1.1 Objectives

The objectives stated in the project's planning report were framed around a haptic master–slave system with physical springs at the master and virtual spring impedance at the slave. The delivered system diverged from that plan in several respects, for reasons set out in [Chapter 6](#) and analysed in full in the accompanying divergence record. The objectives below are the revised set, aligned to what was built and measured.

1. To design and fabricate a miniature instrumented mannequin master replicating the Gen3 morphology at reduced scale, and to characterise its sensing.
2. To derive and implement the mapping from master pose to slave end-effector command, including the

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coordinate-convention conversion required between the manufacturer’s controller and the robot middle-ware.

3. To implement a safety architecture appropriate to a manipulator mounted beside a person, and to validate it by systematic fault injection rather than by inspection.
4. To characterise the reachable workspace of the mounted configuration and to determine which manipulation tasks it can support.
5. To establish a verification methodology yielding traceable evidence for every claim, and to apply it across the task set.
6. To identify, by measurement, the constraints that must be resolved before hardware validation and a user study become possible.

1.2 Challenges

Four difficulties consumed the majority of the effort, and each is treated at length in the chapters that follow.

The first was *kinematic dissimilarity between master and slave*. The master replicates the Gen3’s joint sequence but not its proportions: its links sum to 0.272 m against the full-scale arm’s 0.902 m reach. A joint-to-joint mapping is therefore unavailable, and a mapping in task space must contend with the fact that the two chains reach different sets of poses. The resolution adopted was to abandon joint correspondence entirely and map displacement of the master’s tip onto displacement of the slave’s end effector about a re-latched anchor, which is described in [Chapter 3](#).

The second was *a coordinate-convention mismatch*. The manufacturer’s controller reports joint positions in degrees over $[0, 360)$ while the middleware uses radians over $(-\pi, \pi]$, and the two conventions must be reconciled on every command. The failure mode is not a small error: an angle of 350° interpreted as $+350$ rather than -10 commands very nearly a full revolution of a joint standing next to a person. The conversion was therefore isolated in a single module and tested at the seams where such mistakes occur.

The third was *calibration without power-on discontinuities*. The potentiometers report absolute values whose meaning depends on how the master was assembled, so a zero reference must be captured and stored. If that reference is taken from a frame that has failed validation, or if the commanded pose is referenced to the origin of the master’s own frame rather than to the operator’s actual hand position, the first command issued after start-up is a step of the master’s entire tip vector.

The fourth, and the one that proved most pervasive, was that *almost every failure in this system is silent*. An arm that has stopped moving looks identical whether it is blocked for a good reason, blocked for a bad one, or receiving no data at all. A substantial part of the engineering effort went into making internal state legible from outside, and a recurring finding of the results is that the measuring instruments failed more often, and more quietly, than the system being measured.

1.3 Contributions

The contributions claimed are limited to those the implementation and its recorded artefacts can support. Each is stated with its maturity: whether it was implemented, verified in simulation, or validated on physical hardware. **No contribution in this thesis was validated on physical hardware**, for reasons given in [Chapter 6](#).

1. **A complete, safety-instrumented teleoperation stack** for a backpack-mounted pair of Gen3 arms, comprising master sensing, a spherical pose decomposition, collision-aware inverse kinematics with redundancy resolution, a geometric clearance margin referenced to the wearer’s body, and a multi-path emergency stop whose blocking behaviour is externally observable. *Implemented; verified in simulation.*
2. **A coordinate-convention conversion layer** that isolates the degree-to-radian and wrapping problem in one auditable place, with a shortest-path joint difference that distinguishes continuous joints from limited ones. *Implemented; unit-tested.*
3. **A workspace characterisation of the mounted configuration**, including the finding that the two arms share no reachable workspace under the as-built mount, which blocks inter-arm handover for geometric reasons that no control or planning change can address, together with a quantified and buildable mount modification that would resolve it. *Verified in simulation; Chapter 5.*
4. **A quantified account of degraded-input teleoperation** — which capabilities survive the loss of individual sensing channels, what each loss costs, and evidence of progressive hardware degradation between two dated recordings. *Measured from recorded hardware data.*
5. **A verification methodology** requiring that an instrument recover a known answer before a surprising measurement taken with it is reported, applied across a set of recorded runs covering every task, scenario and autonomy condition with automated checking of object handling, gripper actuation and wearer clearance. *Applied throughout.*

The fifth of these arose from necessity rather than principle. Across the project, unexpected results proved more often to be defects in measurement than in the system, and the discipline of validating an instrument before believing its output caught a substantial number of errors that would otherwise have entered the record. It is, in the author’s assessment, the most transferable outcome of the work.

1.4 Thesis structure

[Chapter 2](#) reviews supernumerary limb design, master–slave teleoperation and the mapping problem between dissimilar morphologies. [Chapter 3](#) describes the system as built, from the master hardware through to the safety architecture. [Chapter 4](#) reports the measured outcomes, organised by validation level and labelled throughout with the strength of evidence behind each. [Chapter 5](#) presents the workspace characterisation separately, because it is a chain of measurements whose consequences determine the shape of the task set. [Chapter 6](#) interprets these results, compares the delivered system against the planning report’s targets, and states the limitations honestly. [Chapter 7](#) restates what was achieved against the objectives and sets out the work that would most improve the system.

Chapter 2

Background

2.1 Supernumerary robotic limbs

Supernumerary robotic limbs (SRLs) augment rather than restore: the wearer keeps their own limbs and gains additional ones [TL21, P⁺21]. The design space is broad. Early systems were oriented towards bracing and body support, in which the extra limbs take load rather than manipulate [PA16]; later work added active impedance control to improve the safety and naturalness of contact [Z⁺23]. Recent reviews quantify the field’s design parameters, actuation choices, intent-recognition strategies and feedback modalities [YHC⁺21, Z⁺26].

Two questions recur, and both bear directly on this project. The first is how the extra limbs are *commanded*, given that the wearer’s own limbs are generally occupied by the primary task. The second is whether the wearer comes to experience the limbs as part of their own body, which is normally framed in terms of body schema and ownership [PLF⁺23].

The configuration used here answers the first question by placing the commanding person outside the wearer’s body entirely, and renders the second inapplicable: the arms are never mechanically coupled to the wearer’s limbs, so ownership language does not apply to them. The MUVE platform on which the work was conducted mounts two Kinova Gen3 arms on a backpack frame [Imp, Kin22].

2.2 Teleoperation with a separate wearer

The direct predecessor to the two-person configuration is Fusion, which mounted a three-axis robotic head with stereo vision and two six-degree-of-freedom arms on a backpack worn by a *surrogate*, and drove them from a remote operator wearing a head-mounted display [SSM⁺18]. Fusion defined three levels of what it termed bodily driven communication — *Direct*, *Induced* and *Enforced* — distinguished by how far the robot arms were physically coupled to the surrogate’s own arms.

It is worth being precise about what those levels are, because the naming invites a misreading. They are levels of *physical coupling*, not levels of machine autonomy: in all three the operator commanded every motion and the machine decided nothing. Fusion was moreover presented as a two-page Emerging Technologies demonstration and reported no user study and no quantitative evaluation. It therefore establishes the architecture as buildable and compelling, rather than as characterised.

The spatial behaviour of back-mounted arms around a wearer has more recently been studied directly, with a concealed separate operator and varying apparent autonomy [ZGD⁺26]. That work reported that higher autonomy did *not* improve the wearer’s perceived safety, and produced body-zone guidance on approach distances and

trajectory shapes. It is relevant here both as a precedent for the two-person arrangement and as a caution against assuming that autonomy is experienced as reassuring.

2.3 Master–slave teleoperation and the mapping problem

Bilateral teleoperation, in which position and force information flow in both directions, is the classical route to transparent interaction, and impedance control at the follower is the usual mechanism [dWKB24]. Where the leader and follower have *dissimilar* morphologies, a joint-to-joint mapping is unavailable and the correspondence must be established in task space, at the cost of a scaling calibration and a residual mapping error.

Physical prototyping with miniature models is an established route to validating such a mapping before committing to full-scale hardware, and miniature masters have been shown to provide useful inherent haptic feedback [GA21]. The device built for this project is of that family: a scaled replica of the follower’s own kinematic chain, so that the operator’s intention maps onto robot motion without an intervening abstraction.

2.4 Compliance and series elastic actuation

Series elastic actuators place a compliant element between the motor and the load, trading positioning bandwidth for force fidelity, shock tolerance and safety in contact [TGT⁺22, LKK19]. The planning report for this project proposed a dual-layer compliance strategy: physical springs at the miniature master and virtual spring impedance at the full-scale follower, the latter through the manufacturer’s impedance mode. Chapter 6 explains why the impedance layer was not delivered, and what constraint prevented it.

2.5 Teleoperation from a moving base

Where the base of a manipulator is not inertial, base motion enters the control problem as a disturbance. For supernumerary limbs the base is the wearer’s own body, and the disturbance is their postural sway. Zhang et al. model supernumerary arms on a floating base and report tracking errors of (1.37 ± 0.58) mm with the human shoulder as that base [Z⁺24].

The limitation of that result for the present work is that the wearer is also the operator, so the disturbance is *self-generated* and the person’s own motor commands are available as feed-forward. Separating the roles removes that information entirely, which is the distinction this project is positioned against.

2.6 Simulation

The planning report proposed Isaac Lab for offline tuning of impedance parameters [NVI26]. The delivered work used MuJoCo for early dynamics and ROS 2 with MoveIt 2 for the final collision-aware kinematics; Chapter 6 gives the reasons.

2.7 Research gap

The gap the planning report identified was the absence of studies combining a custom miniature dual-arm master with direct teleoperation of a wearable dual-arm platform. That remains broadly true, but it requires two

corrections which this thesis makes explicitly.

First, Fusion already established the operator/wearer architecture [SSM⁺18], so novelty cannot be claimed for the architecture itself — only for its characterisation, which Fusion did not attempt. Second, the question of how autonomy affects the wearer is *not* untouched: [ZGD⁺26] measured exactly that and found autonomy made perceived safety worse. What remains genuinely open is the conjunction: no study has measured, in a system where one person wears the arms and another drives them, whether machine autonomy changes task performance *and* the wearer’s experience at the same time. Those two literatures currently point in opposite directions, and nobody has measured both ends in one system.

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Chapter 3

System Design and Implementation

3.1 Architecture

The system is organised as eight project packages under ROS 2 Jazzy, running on Ubuntu 24.04 under WSL2. The dependency direction is one-way and deliberate: `srl_teleop` contains teleoperation only and depends on nothing else in the workspace, `srl_perception` adds detection, `srl_autonomy` depends on both, and `srl_experiments` depends on all of them.

That direction is load-bearing rather than tidy. Direct teleoperation is the baseline against which every autonomy condition is compared, so if the teleoperation package imported from the autonomy package the comparison would be circular. The property was verified two ways: by search, and by physically removing the autonomy and perception packages from the install tree and confirming that the follower still reached its running state.

3.2 The master arm

The operator commands the system through a miniature mannequin: a scaled replica of the Gen3 kinematic chain instrumented with single-turn potentiometers and an inertial measurement unit, sampled by a Teensy 4.1 and streamed over USB serial. Link lengths are $\ell = [43, 37, 43, 37, 43, 36, 33]$ mm (`master_calibration.py`), summing to a full extension of $L_m = 0.272$ m against the Gen3's 0.902 m reach, a size ratio of 3.32. Joints alternate roll and bend, matching the Gen3 morphology.

3.2.1 Serial protocol

Frames are newline-terminated ASCII at 115 200 baud:

Listing 3.1: Master frame format, one line per sample.

```
1 k1j1:<deg> ... k1j7:<deg> K1IMU:ax,ay,az,gx,gy,gz
2 k2j1:<deg> ... k2j7:<deg> K2IMU:ax,ay,az,gx,gy,gz
3 fsr1:<n>,fsr2:<n>,btn1:<0|1>,btn2:<0|1>
```

Frames arrive at approximately 50 Hz but the potentiometers update at only 14.7 to 17 Hz, so 82 to 88 % of adjacent frames are bit-identical. That aliasing is not a curiosity: any statistic computed over consecutive rows is dominated by duplicates, and it invalidated an earlier noise-floor measurement ([Chapter 4](#)).

The port is discovered rather than configured, because the board enumerates as either `/dev/ttyACM0` or `/dev/ttyACM1` depending on attachment order. Access is made exclusive with an advisory lock on the open

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descriptor. This is not merely defensive: the port *search* itself reads bytes, so probing a port another node owns removes frames from that node’s stream.

3.2.2 Pose decomposition

The commanded position is built in spherical form about the master’s shoulder rather than by forward kinematics of the full chain, because the full solution collapsed directionally when evaluated against recorded gestures. Writing e for elevation, a for azimuth and r for reach,

$$\mathbf{p}_m = r(\cos e \cos a, \cos e \sin a, \sin e), \quad (3.1)$$

with elevation taken from gravity alone,

$$e = \arcsin(\hat{\mathbf{a}} \cdot \hat{\mathbf{u}}), \quad \hat{\mathbf{u}} = \frac{\mathbf{a}_{\text{meas}}}{\|\mathbf{a}_{\text{meas}}\|}, \quad (3.2)$$

where $\hat{\mathbf{a}}$ is the arm’s long axis in the sensor frame, captured once with the arm hanging, and $\mathbf{a}_{\text{meas}}$ is the measured specific force. Because an accelerometer at rest reports specific force pointing upward, $\hat{\mathbf{u}}$ is world-up expressed in the sensor frame, and Eq. (3.2) needs no mount rotation and cannot drift. Azimuth a comes from the shoulder roll potentiometer alone, zero-referenced; reach r is the magnitude of the forward kinematics of the proximal joints.

3.2.3 A convention error worth recording

The specification called for both $\hat{\mathbf{a}} = -\hat{\mathbf{a}}_{\text{rest}}$ and a leading negation in Eq. (3.2). The two compound: together they place the hanging rest pose at $+90^\circ$, i.e. straight up. Since $\cos$ is even and only $\sin$ carries the sign, the effect is to mirror the z axis alone — the symptom being that “down” commanded a higher pose than “up”. Only one negation is correct, and the rest pose is asserted at capture time as a self-check.

3.3 Master-to-slave mapping

The commanded end-effector position is a scaled displacement about a latched reference:

$$\mathbf{p}_{\text{cmd}} = \mathbf{p}_{\text{anchor}} + s(\mathbf{p}_m - \mathbf{p}_{\text{ref}}), \quad (3.3)$$

where s is the per-arm scale and $\mathbf{p}_{\text{ref}}$ is latched on the first valid frame and re-latched on every clutch engagement. The scale is 1.0, chosen from measured reachability rather than from geometry: at unity the commanded span is 0.44 m with 92 % and 95 % of recorded poses solvable for the left and right arms, and a one-to-one mapping is the most predictable for an operator.

Latching the reference matters more than it appears to. An earlier version referenced the master’s own frame origin, so the first command after start-up was a step of the master’s entire tip vector, and the operator had to hold a calibrated neutral pose for the system to be safe to start.

3.4 Coordinate conventions

The manufacturer’s controller reports joint positions in degrees over $[0, 360)$; the middleware uses radians over $(-\pi, \pi]$. Writing g for the conversion,

$$\text{wrap}_{180}(d) = 180 - ((180 - d) \bmod 360), \quad (3.4)$$

$$\text{wrap}_{\pi}(\theta) = \pi - ((\pi - \theta) \bmod 2\pi), \quad (3.5)$$

and the ordering is the substance: the controller value is wrapped into $(-180, 180]$ and then converted, never the reverse. Motion between two poses uses a signed shortest difference, $\Delta\theta = \text{wrap}_{\pi}(\theta_t - \theta_c)$, applied only to the continuous joints $\{1, 3, 5, 7\}$; the limited joints use plain subtraction, because for them wrapping would mean driving through a hard stop.

The failure mode justifies the isolation: an angle of 350° read as $+350$ rather than -10 commands very nearly a full revolution of a joint standing beside a person. The conversion is therefore confined to one module, `kortex_convention.py`, and unit-tested at the seams.

3.5 Calibration

Three quantities are captured and stored per arm. The *potentiometer zero* is a per-channel offset and sign, applied as a relative reference so the reading has meaning independent of assembly. The *IMU reference* $\hat{\mathbf{a}}$ is captured once with the arm hanging. The *gyroscope bias* is the mean of a settled rest sample, flagged stale if the residual rate exceeds $1^\circ/\text{s}$.

A trap worth recording: the bias must be captured from a settled stream. Reopening the serial port resets the microcontroller, and the IMU emits nonsense while settling; a bias captured that way read $27.9^\circ/\text{s}$ of noise where the settled figure was $0.11^\circ/\text{s}$.

The *workspace anchor* is the arm’s end-effector pose at its home configuration, read from the transform tree rather than assumed. Because the anchor is derived from the home pose, any change to the mount or to the home angles invalidates it.

3.6 Motion generation and safety

Commanded poses are solved with TRAC-IK through MoveIt’s inverse-kinematics service with collision checking enabled against a model that includes the wearer. Two mechanisms handle failure. On an unsolved pose the redundant degree of freedom is re-seeded and the solve retried, fanning the elbow either side of its natural posture. On a solution far from the current state the follower commits a bounded step towards it rather than discarding it:

$$\mathbf{q}_{k+1} = \mathbf{q}_k + \text{clip}(\mathbf{q}^* - \mathbf{q}_k, -\Delta_{\max}, +\Delta_{\max}), \quad (3.6)$$

which converges in $\lceil d/\Delta_{\max} \rceil$ cycles. The distinction is not cosmetic: an earlier implementation *discarded* any solution further than the threshold, which can never converge, because a discarded solution leaves the arm where it was and the next solution is equally far away.

3.6.1 Clearance

Separately from the binary collision check, a geometric clearance is computed from the transform tree between the arm's distal links and primitive approximations of the wearer's body, giving a margin rather than a contact flag. The follower refuses to publish below a configurable floor: 0.05 m in simulation and 0.12 m in the real-robot configuration. [Chapter 5](#) shows that this distinction — contact versus margin — has substantial consequences.

3.6.2 Emergency stop and dead-man

Three independent triggers latch until an explicit reset: a service call, both master buttons held simultaneously, and a dead-man on the master stream. On trip, each arm's measured joint positions are republished as a short trajectory, so a follower that continues commanding is continuously overridden.

The dead-man keys on the *source timestamp* of the master pose, not on message arrival, and this is the single most important detail in the safety layer. The master node does not fall silent when it degrades: it substitutes its last good value and continues publishing at full rate. An arrival-keyed dead-man therefore cannot fire on the failure that actually occurs, and only a clean cable removal produces silence — which is why every cable-pull test had passed.

3.7 Control modes

3.8 Verification methodology

Because much of this work consists of measurements, the method by which a measurement is trusted is itself part of the method. Two rules were adopted after repeated failures.

First, *a pose that passes a single inverse-kinematics call is not a reachable pose*. TRAC-IK restarts randomly, so a pose at the edge of the feasible set is a coin flip. One pose measured 72 to 82 % feasible per call, which passes a five-repeat check roughly one time in five. Verification is therefore $N = 10$ repeats over a path densified to 20 mm, because the arm flies the segments between declared waypoints and those are not reachable by virtue of their endpoints being reachable.

Second, *an instrument must recover a known answer before a surprising measurement taken with it is reported*. Every analysis routine that will touch experimental data has a test that feeds it an input whose answer is known by construction.

Chapter 4

Results

4.1 How to read this chapter

Every result is labelled with the strength of evidence behind it: **Simulated** (measured against a running planning and control stack with a mock hardware interface), **Mock-exercised** (the code path ran and its failure modes were injected, but no physical device was involved), or **Hardware** (a physical arm, microcontroller or participant took part).

No result in this chapter is of the Hardware tier for the robot. No experiment was run with a participant and no logged artefact from a physical Kinova arm exists. One category of hardware evidence does exist and is reported in [Section 4.3](#): recorded master-arm data.

4.2 The five-task set

Nine task families were consolidated to five. The consolidation criterion for the bimanual tasks was strict: one arm alone must be unable to complete the task *given unlimited time*. Because the two reachable sets are disjoint ([Chapter 5](#)), that can only be achieved in two ways — a rigid object spanning the dead band, or two targets requiring simultaneous satisfaction — and both are used.

4.3 Master-arm channel health and degradation

Twelve recordings made on 27 July 2026 constitute the only physical-hardware data in the project: 10 890 rows at 100.0 Hz over 109 s, capturing all fourteen potentiometer channels, two inertial units, two force-sensitive resistors and two buttons, in six labelled directions per arm. (*Hardware.*)

Classifying channels between *distinct* sensor updates rather than between rows — the distinction matters, as [Section 4.7](#) explains — gives five live channels on the left arm and two on the right at that date. Re-measuring on 6 August against the same hardware shows substantial degradation ([Fig. 4.1](#)): the left arm’s third and fourth channels, which spanned 234° and 175° with no dropouts in July, are classed incoherent in August.

4.3.1 What degraded input costs

Replaying recorded traces with every coherent channel held at its median isolates the contribution of the failed channels alone — a perfectly steady hand, so every millimetre of output is contributed by the faults. The left

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Table 4.1: The consolidated task set. Every coordinate was verified at $N = 10$ repeats over the full densified path with both arms confirmed at the home pose; total failures zero.

Task	Arms	Why it earns its place
1 Positioning	L, R, both	Fitts characterisation and warm-up; the single-versus-both contrast is the attention bottleneck, needing no object to exist
2 Pick and place	one at a time	Single-arm manipulation baseline and the only discrete success/failure outcome in the set
3 Rigid carry	both	A rigid body held at two points 500 mm apart spans the dead band; one arm cannot hold two separated points
4 Compliant carry	both	Identical paths, compliant object; failure is nonlinear and abrupt rather than proportional
5 Dual pursuit	both	The two targets lie in disjoint sets, so one arm can never reach both; simultaneity is the success criterion

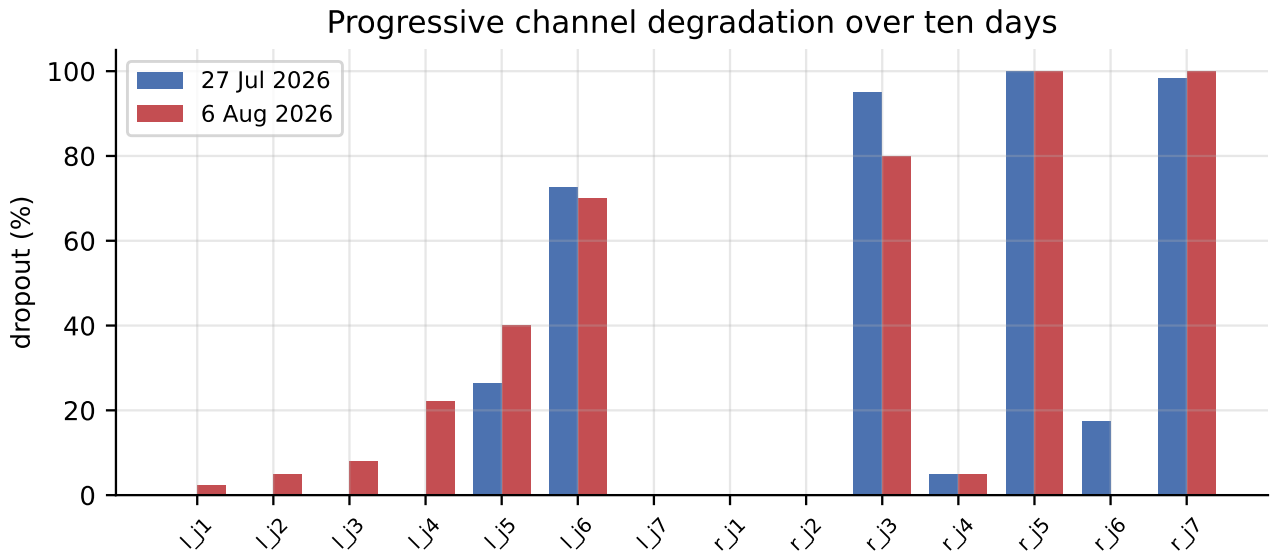


Figure 4.1: Dropout fraction per channel on two dates ten days apart, same hardware. The left arm’s shoulder and elbow channels degrade from fully live to incoherent.

2 % intermittency threshold

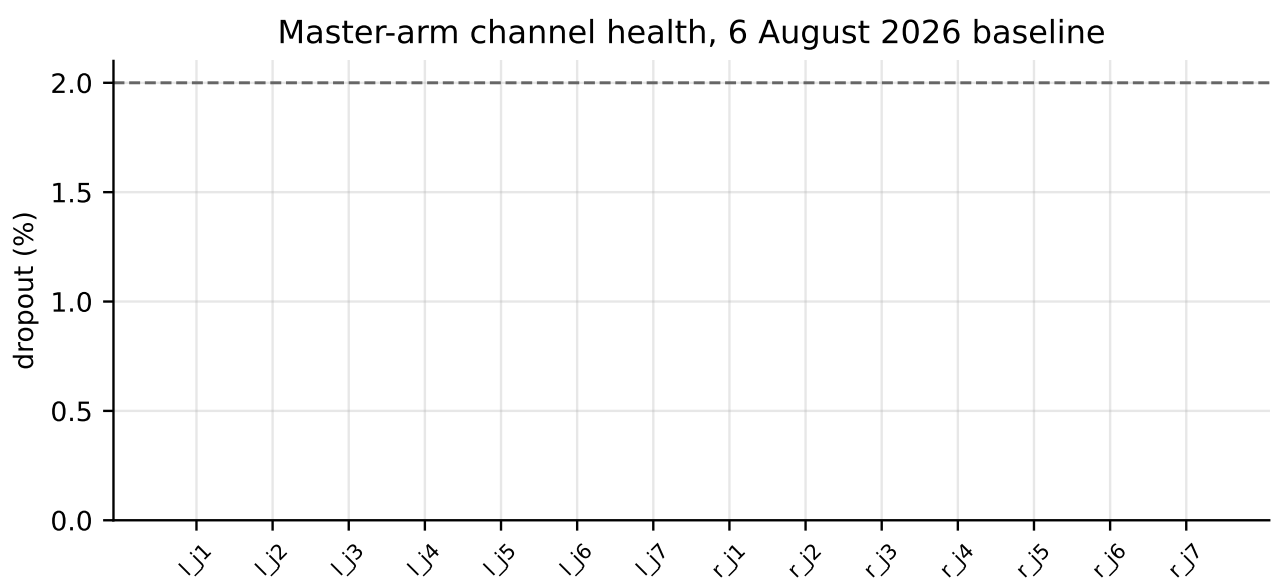


Figure 4.2: The 6 August baseline. Six of fourteen channels are usable.

arm's incoherent channels contribute 54.5 mm of commanded tip motion; the right arm's contribute 0.014 mm. (*Simulated replay of hardware data.*)

Regressing true reach on the surviving channels shows the left arm retains *no* reach observable once its shoulder and elbow channels are excluded ($R^2 = 0.133$, residual 93 % of the true spread), whereas the right arm does ($R^2 = 0.986$). This identifies the two specific channels whose repair would restore the most capability, and it is the highest priority hardware action.

4.4 Safety layer

Fourteen distinct faults were injected and all fourteen were handled, meaning detected, responded to in the designed way, and made visible in the published state. (*Mock-exercised.*) Faults injected mid-trial into the experiment runners caused the affected trial to be marked invalid with the cause recorded, the session to continue, and the analysis to exclude the trial as pre-registered.

The most consequential defect found was that the dead-man could not detect the failure that actually occurs, for the reason given in [Chapter 3](#). After keying it on the source timestamp, the measured trip times were 0.94 to 0.99 s for a degraded-but-publishing master and 0.97 s for a silent one, against a 1.00 s deadline, with no false

MEASURE:positive over 243 fresh frames.

4.5 Clearance

Driving every task and measuring clearance produced the result with the largest consequence for the programme, developed fully in [Chapter 5](#). Across all recorded runs, 48 of 87 sit below the 0.12 m real-robot floor while passing the planner's collision check ([Fig. 4.3](#)).

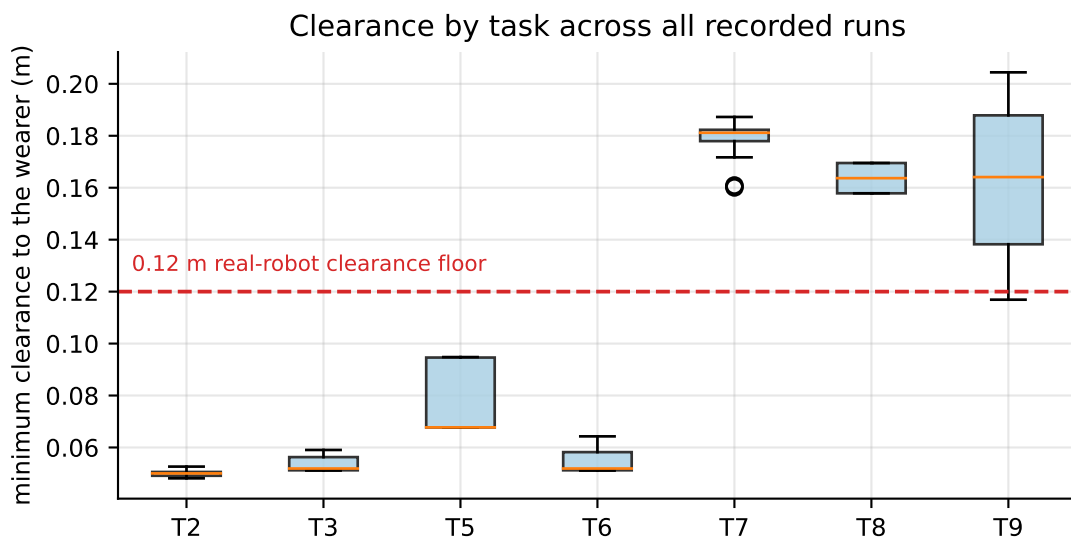


Figure 4.3: Minimum clearance to the wearer across all recorded runs. Tasks working at $|x| \geq 0.35$ m clear the floor comfortably; those in the inboard band do not.

4.6 Recorded verification

Every task, scenario and control condition was driven in simulation and recorded from multiple fixed camera angles, with automated checking of object handling, gripper actuation and wearer clearance. Across all runs no arm link entered the wearer’s collision geometry and no run showed a stationary robot. (*Simulated.*)

4.6.1 A grasp that was not a grasp

The first round of recordings contained a defect that no automated check caught and that only viewing the clips revealed: the gripper was never commanded. The mock hardware interface initialises the driven knuckle at a fully closed value, and because nothing in the recording path issued a gripper command, the hand remained shut throughout while the grasp was represented solely by an attached marker.

The object-attachment check passed, and correctly so: the marker genuinely was following the gripper. It was following a gripper that never opened. This is the clearest instance in the project of a verification that was true and useless.

The recording path now commands the gripper controller directly and executes the full sequence — pre-grasp standoff along the approach vector with the wrist aligned to the object, straight-line approach, closure to the object’s own width rather than to full closure, lift back along the vector — using the existing grasp-pose generator, with the marker attaching only once the *measured* joint indicates closure.

4.6.2 Where a top-down grasp is refused

Exercising the generator produced a quantified refusal that is itself a result. A top-down grasp requires $|x| \geq 0.30$ m: zero of nine candidates solve at 0.25 m and nine of nine at 0.30 m, and the failure persists with collision checking disabled, so it is kinematic infeasibility of the wrist orientation rather than obstruction. Grasps at the container, at the tray edge and at the tool cradle are therefore refused, and the recordings show the arm doing what it can with the refusal displayed rather than pretending otherwise.

4.7 The instrument failed more often than the system

A finding that shaped the methodology more than any design decision is that surprising measurements were more often defects in the measuring instrument than in the system measured. The instances are numerous enough to be a result rather than an anecdote.

A noise floor of 0.000 on every sensor channel was the recorder oversampling a 15 Hz sensor at 50 Hz, so 82 to 88 % of adjacent rows were bit-identical. A workspace statistic varying by a factor of two between identical runs was a summary statistic sensitive to a rare tail event; on medians the inter-quartile range is 0.000 m. A set of verification clips reported as failing were failing a colour test calibrated against the requested marker colour rather than the rendered one — eleven clips whose objects had just been inspected by eye. A carried object reported as stationary was a centroid dragged towards fixed overlay text, of which 529 of 587 matching pixels were letters. In each case the system was behaving correctly and the instrument was not.

4.8 Unit tests

Twenty-eight test files are present, including known-answer recovery for every continuous task metric, the discrete task outcome, the inverse-kinematics guard convergence and the cross-arm interference fit. Style tests fail and failed prior to this work, because the codebase uses double quotes throughout; this is pre-existing and is not a regression.

Chapter 5

Workspace Characterisation and its Consequences

This chapter is separated from [Chapter 4](#) because it is not a single measurement but a chain of them, and because its consequences determined the shape of the task set. It reports a negative finding together with a quantified and buildable mechanical fix; the finding alone would be a complaint rather than a contribution.

5.1 Continuous reachability from the home pose

Reachability was measured as a swept extent rather than as pointwise feasibility, because teleoperation requires the arm to travel to a pose from where it already is, without a branch change. Twenty-six directions were swept in 1 cm steps from the home configuration, each step seeded from the previous solution exactly as the follower does.

Repeating the sweep ten times gave identical quartiles: the inter-quartile range was 0.000 m in every direction for both arms. The measurement repeats. What did *not* repeat in earlier runs was the summary statistic: isotropy taken as the minimum over directions of a single sweep is dominated by rare early terminations and varied by a factor of two between runs. On medians it is stable, at 0.16 for the left arm and 0.10 for the right ([Fig. 5.1](#)).

The early terminations were traced to the measuring tool rather than to the robot. A sweep is a chain of up to ninety solver calls and any single failure ends that direction; the follower does not suffer this because it re-seeds the redundant joint and retries. Giving the sweep the same retry reduced early terminations from nine to zero on the left arm and three to zero on the right, while the medians barely moved — which is what establishes that the outliers were chain compounding rather than geometry.

5.2 The two arms share no reachable workspace

Over a grid of sixty-three cells spanning the frontal working volume, with the arms verified at the home pose, **no cell was reachable by both arms**. The result survives the two obvious objections, as [Table 5.1](#) shows: freeing the wrist yaw raises each arm’s individual reach from sixteen–eighteen cells to twenty-one and still yields no shared cell, and repeating the test with more redundancy retries changes nothing.

A fine sweep in 0.05 m steps across the plane on which the arms would have to meet locates the boundary precisely: the right arm reaches $x \leq -0.10$ m, the left reaches $x \geq 0.15$ m, and there is a 0.25 m dead band between them in which neither arm can place its end effector ([Fig. 5.2](#)).

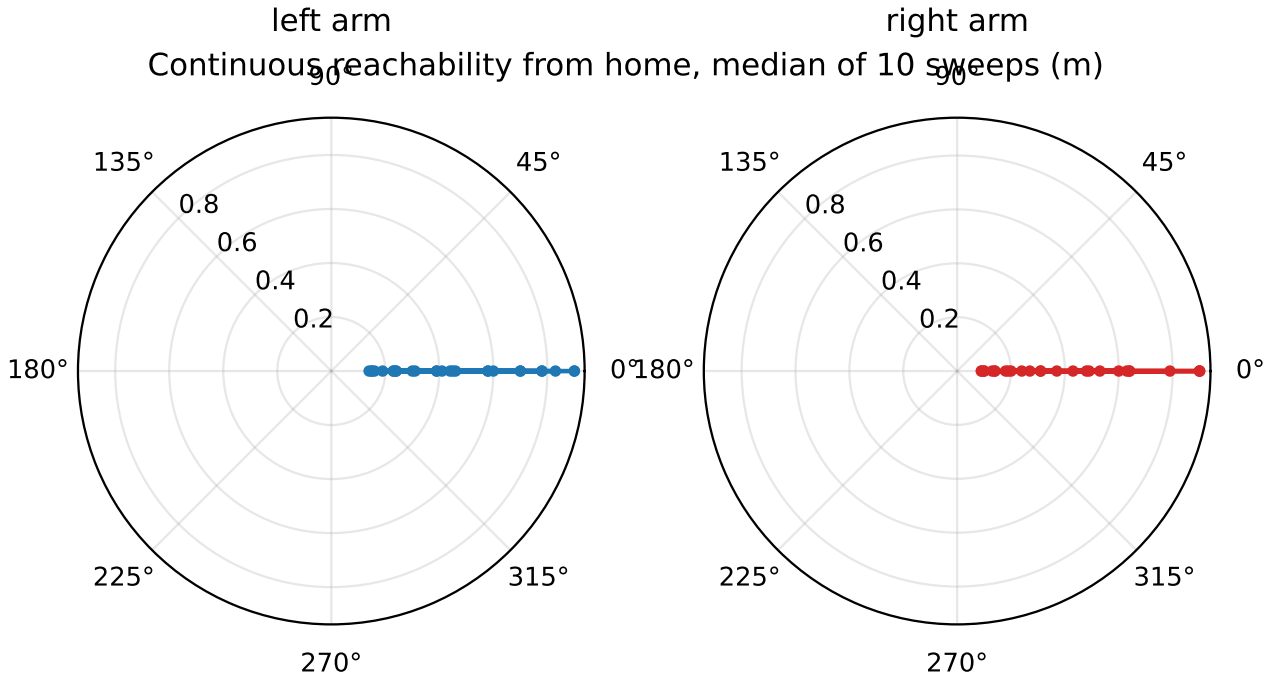


Figure 5.1: Continuous reachability from the home pose, median of ten sweeps, projected onto the horizontal plane. The envelope is markedly anisotropic: the median reach is 0.34 to 0.40 m but the worst direction is 0.14 m on the left arm and 0.09 m on the right.

Table 5.1: Both-arm overlap over the 63-cell frontal grid, arms verified at home. The result is robust to wrist freedom and to additional inverse-kinematics retries.

Test condition	Both	Left	Right
Fixed orientation, 3 retries	0/63	16	18
Fixed orientation, 6 retries, 3 repeats	0/63	17	18
Yaw free, 6 retries	0/63	21	21

The two reachable sets do not meet (measured, N=3, yaw-free)

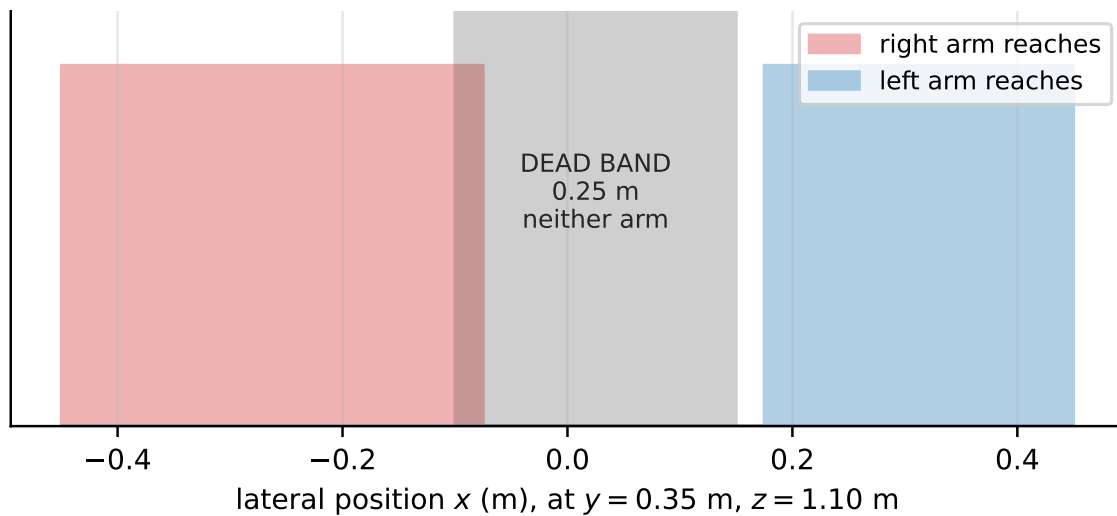


Figure 5.2: The two reachable sets do not meet. Measured at $y = 0.35$ m, $z = 1.10$ m with yaw free and three repeats per point.

5.2.1 What this does and does not establish

It establishes that, within $x \in [-0.6, 0.6]$, $y \in [0, 0.4]$, $z \in [0.90, 1.30]$ m, with the as-built mount and the legacy home joint angles, the two reachable sets do not intersect.

It does *not* establish behaviour outside that volume, nor does it generalise to other mounts. The correct claim is that the arms share no workspace *as mounted and as parked*, and the remedy is mechanical.

5.3 The fix is buildable

A parametric sweep over candidate mount placements shows the null result is not intrinsic. Moving the mounts inboard, forward and down, with additional forward tilt, raises the number of shared cells from three to **twenty-two** on the purely kinematic measure, within a buildable 0.20 m mount separation (Fig. 5.3).

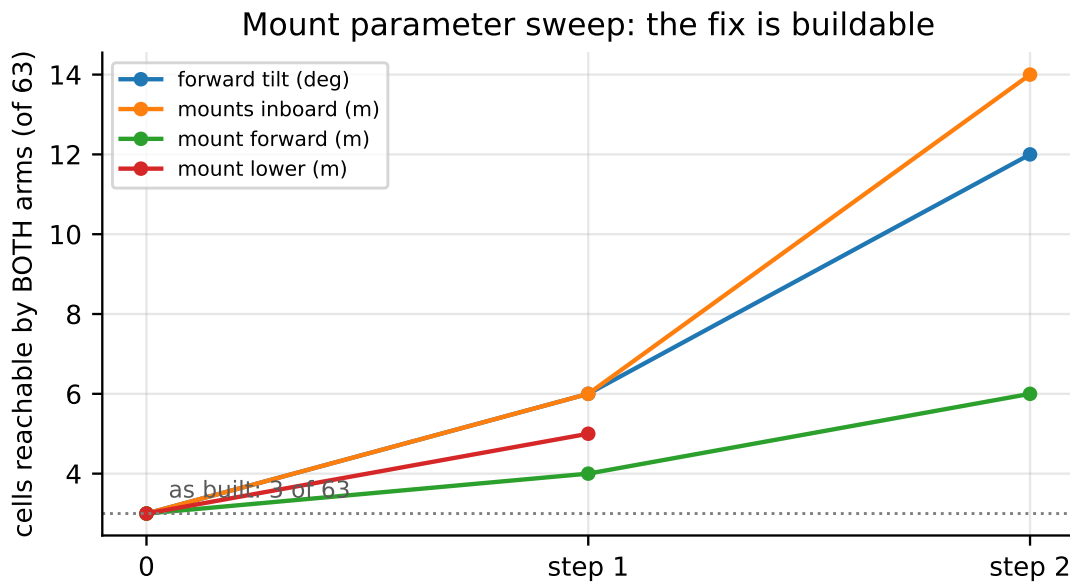


Figure 5.3: Single-parameter sweeps from the as-built mount. Forward tilt and inboard translation both help substantially; splaying the mounts outward *hurts*. The sweep is purely kinematic — it moves the arms’ bases without moving the wearer — so these are upper bounds and any candidate requires a clearance check before adoption.

The sweep was recorded and deliberately not applied, because changing the mount invalidates the home pose, the workspace anchor and every clearance figure reported in [Chapter 4](#).

5.4 One boundary, two independent constraints

Two further measurements, taken for different reasons, land on the same lateral boundary and together make the case that the inboard band is unusable.

The first is wearer clearance. Every scenario in the chest-height band passes inverse kinematics with collision checking enabled — the planner reports the poses valid — and yet the arms pass within 48 to 67 mm of the wearer’s torso, inside the 120 mm margin the real-robot configuration enforces. The reconciliation is that the two tests ask different questions: the planner’s check is binary contact, whereas the floor is a margin, and a pose can

satisfy one while being refused by the other. Sweeping the lateral half-span shows clearance rising monotonically and crossing the floor at 0.25 m.

The second is grasp feasibility. A top-down grasp requires the wrist to adopt an orientation with its approach axis vertical. Sweeping the same lateral coordinate gives a sharp cliff: zero of nine grasp candidates are solvable at $|x| = 0.25$ m and nine of nine at $|x| = 0.30$ m. The failure persists with collision checking disabled, so it is kinematic infeasibility of the wrist orientation rather than obstruction.

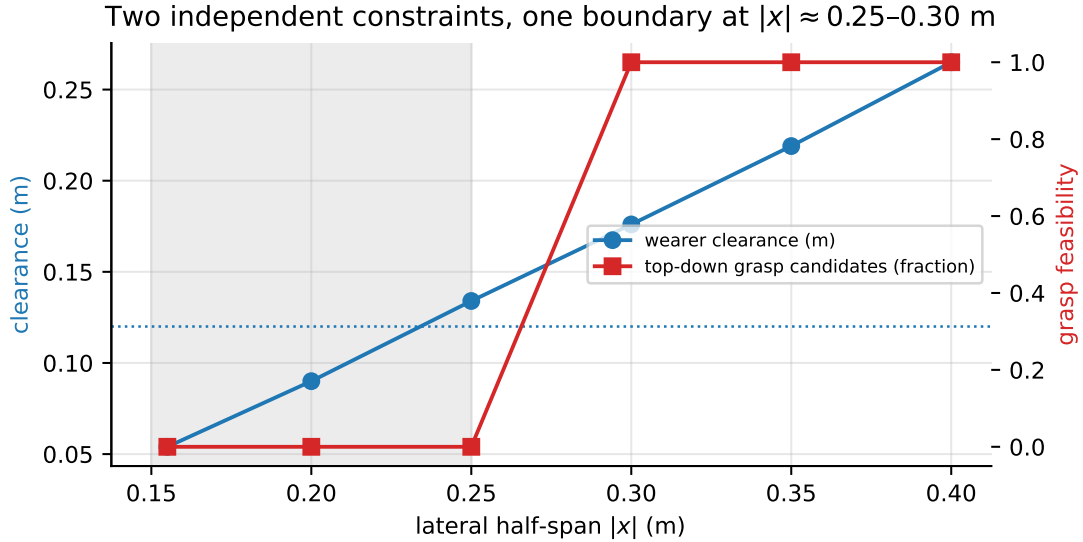


Figure 5.4: Wearer clearance and top-down grasp feasibility against lateral half-span. The two constraints were measured independently and for different purposes, and both place the usable boundary at $|x| \approx 0.25$ to 0.30 m.

5.5 Consequences for the task set

Three consequences follow directly, and each is visible in the task definitions of [Chapter 4](#).

Inter-arm handover is impossible. It requires both arms at one point, and no such point exists. This removes an entire task family, and no control, planning or autonomy change can restore it.

A bimanual task must either span the gap or exploit simultaneity. A rigid object held at two points can bridge the dead band; two targets in the two disjoint sets can require simultaneous satisfaction. Both routes are used.

The coupled objects had to be re-specified. The original 310 mm tray placed each grip at $|x| = 0.155$ m, inside the dead band, and ten of twenty-five waypoints failed once the arms were verified at home. Widening to 500 mm places each grip at $|x| = 0.25$ m, which satisfies the clearance boundary and the reachability boundary simultaneously. The compliant carrier had to lengthen with it: retention of a ball of radius r requires sag of at least $2r$, so

$$\text{sag}(s) = \sqrt{\left(\frac{L}{2}\right)^2 - \left(\frac{s}{2}\right)^2}, \quad s_{\max} = 2\sqrt{\left(\frac{L}{2}\right)^2 - (2r)^2}, \quad (5.1)$$

where s is the gripper separation, L the carrier length and r the ball radius. For $L = 540$ mm and $r = 20$ mm this gives $s_{\max} = 534$ mm, leaving a 34 mm margin at the nominal 500 mm separation ([Fig. 5.5](#)). The length is chosen so that the failure threshold sits *inside* the reachable band: a longer carrier could never fail and would

measure nothing.

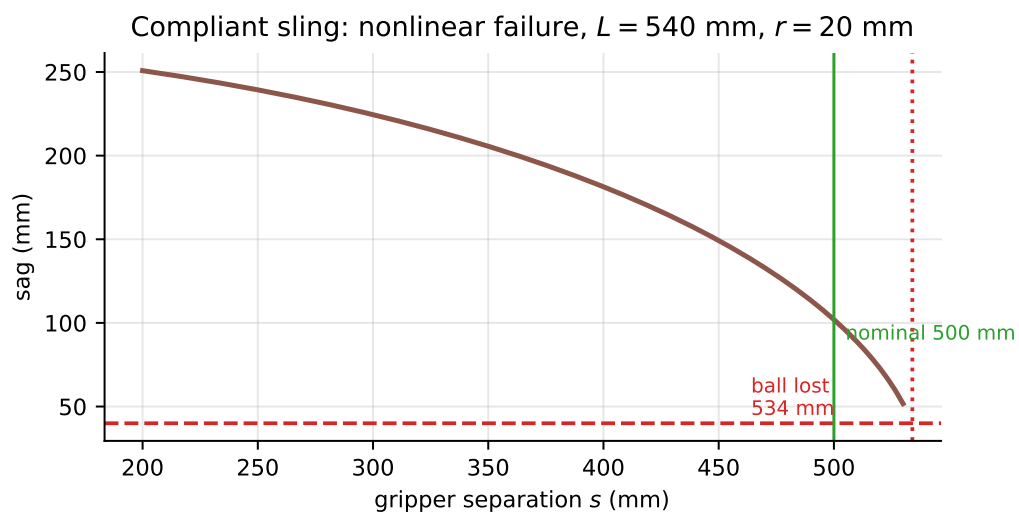


Figure 5.5: Sag against gripper separation for the re-specified compliant carrier. The nonlinearity is the point: sag falls slowly, then the ball is lost abruptly at 534 mm.

Chapter 6

Discussion

6.1 What the geometry decides

The most consequential outcome of this project is not a control result but a geometric one: as mounted and as parked, the two arms share no reachable workspace. That removes an entire class of task, and it does so in a way no planning, control or autonomy change can address, because the constraint is on where the arms can be at all rather than on how they get there.

It is worth stating precisely what follows and what does not. Tasks requiring the two arms to meet at a point are impossible. Tasks requiring two distinct points a fixed distance apart are not, and were shown feasible once the coupled objects were widened to a 500 mm span. The distinction was initially missed, and a bimanual programme was briefly declared blocked on the strength of a test that answered the stronger question. The correction matters more than the original error: the requirement a task actually imposes should be tested, rather than a convenient proxy for it.

That the fix is mechanical and quantified — three to twenty-two shared cells for a buildable mount change — is what turns the null result into a contribution rather than a complaint.

6.2 Two constraints, one boundary

Wearer clearance and top-down grasp feasibility were measured independently, for different reasons, and both place the usable boundary at $|x| \approx 0.25$ to 0.30 m. Neither measurement was designed to corroborate the other, which is what makes the agreement informative: the inboard band is unusable for two unrelated reasons at once.

The clearance half of that pair deserves emphasis because it is easy to mistake for a tuning problem. The planner's collision check and the deployed clearance floor ask different questions — contact against margin — and a pose can satisfy the first while being refused by the second. Attempting to resolve it by moving the working volume forward fails instructively: clearance improves and reachability collapses, because the volume moves towards the edge of the arm's reach as it moves away from the body. The axis along which the conflict resolves is lateral, and that was not obvious in advance. It was found only because the probe recorded *which link* was closest rather than only how close it was.

The ethical position is one-directional. The floor exists to protect a person who did not choose the motion, so the task moves and the floor does not.

6.3 Instrumentation as a first-class concern

The pattern reported in [Section 4.7](#) — that surprising measurements were more often instrument defects than system defects — became the methodological centre of the work. The response, requiring every analysis routine to recover a constructed answer before being trusted with real data, is in the author’s assessment the most transferable outcome.

It is also incomplete, and honesty requires saying so. The rule was adopted part-way through, and the earlier measurements have not all been re-derived under it. Where a figure in this thesis originates in the engineering log rather than in a retained artefact, it is marked as such.

6.4 Silent failure as the dominant mode

The safety work consistently found that the problem was not mechanisms failing to act, but that acting and not acting looked identical from outside. The dead-man defect is the clearest case: the mechanism was correct, its trigger condition was correct, and it could not fire on the failure that actually occurs, because the upstream node degraded by republishing stale data rather than by falling silent. Every test of it had passed, because every test removed the cable.

The general lesson is that a test exercising a failure mode different from the one that occurs provides confidence without providing safety.

6.5 The plan and the delivery

The delivered system diverges substantially from the planning report, and the divergences are engineering decisions rather than omissions.

The simulator changed from Isaac Lab to MuJoCo and then to ROS 2 with MoveIt. Isaac Lab requires a heavyweight GPU stack; the development machine carries 4 GiB of video memory already committed to the perception models. More importantly, the question that actually needed answering was collision-aware kinematics against a wearer model, which MoveIt provides natively.

The compliance layer was not built. The manufacturer’s impedance mode requires the cyclic control interface, and that interface is unusable on this machine: measured write times of 3 to 6 s against a 10 ms budget, because the realtime channel is UDP and the network translation layer breaks it. The delivered path uses the high-level velocity API instead, which cannot express joint impedance. This is an infrastructure constraint, not a design preference, and it is the reason the haptic objectives could not be attempted.

The master arms are passive. This follows from the above: with no force channel there is nothing for master actuators to render.

6.6 Evaluation against the planning report’s targets

Of the nine targets in the planning report, none was met and none refuted. Four are measurable with what exists; three require building a subsystem first; one requires participants.

The three that require building are worth distinguishing carefully, because calling them merely “unmeasured” would understate the position. Force reflection and closed-loop overshoot are **structurally undefined on this**

platform: the mock hardware interface returns identically zero for every force and effort field, so a number derived from it exists but carries no information. Angular tracking is undefined for a different reason: the commanded orientation is pinned to the anchor and never follows the master's wrist, because the wrist channels are dead or incoherent.

6.7 Limitations

6.7.1 Nothing was validated on the robot

No result here was obtained from a physical Kinova arm or a participant. The simulation uses a mock interface that echoes commands with no dynamics, so it cannot answer questions about payload, compliance, latency under load, or any consequence of real actuator behaviour. Where it has been asked such a question it returned a null result that is a property of the mock rather than evidence about the robot.

6.7.2 The input is degraded and the baseline depends on it

Seven of fourteen master channels are unreliable, and the degradation is progressive: channels healthy in July were incoherent ten days later. Every direct-teleoperation condition depends on those channels, so the baseline against which autonomy would be compared is currently compromised. Repairing two specific channels was identified by regression as the highest-value intervention.

6.7.3 The lateral axis is wrong and cannot be fixed by remapping

Azimuth is derived from the shoulder roll joint alone. That joint's axis is fixed in the mount, so rolling it moves the tip only once a downstream joint has offset the tip from the roll axis; azimuth per degree of roll therefore depends on how bent the arm is. Azimuth and elevation are consequently not independent, which is precisely what the spherical model assumes. No axis permutation can repair this — it is a modelling error, not a sign error — and a hybrid using the full forward kinematics was tested and made the reversal metric worse, because it inherits errors from four suspect joint angles rather than one.

6.7.4 The latency figure, when it exists, is a lower bound

The end-to-end latency measurement designed for this system instruments the control path from serial read to commanded joint state. Under the mock interface the final stage measures command echo rather than physical motion, so the resulting figure is **control-path latency and a lower bound** on true end-to-end latency. That caveat travels with the number wherever it appears.

6.7.5 The experimental programme is designed but unexecuted

Protocols, measures, counterbalancing and analysis routines exist and have been exercised with scripted input.

CONFIRM: No participant has taken part, so every hypothesis remains pre-registered intent.

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6.7.6 The version history cannot date the work

The repository's commit history consists of one large import and a single day of subsequent commits, so it cannot support claims about the order or timing of development. The dated engineering log is the usable chronology and is a laboratory notebook rather than an independent record.

Chapter 7

Conclusion

This project produced a complete, safety-instrumented teleoperation stack for a backpack-mounted pair of seven-degree-of-freedom arms commanded through a miniature mannequin master, together with a characterisation of what that configuration can and cannot do.

Against the objectives of [Chapter 1](#): the master was designed and fabricated and its sensing characterised, including a dated record of progressive channel degradation; the mapping was derived and implemented, including the coordinate-convention layer that isolates a class of error whose failure mode is a near-full joint revolution; the safety architecture was implemented and validated by fault injection rather than inspection; the workspace was characterised, yielding the finding that determines the task set; a verification methodology was established and applied across the recorded runs; and the constraints blocking hardware validation were identified by measurement rather than assumed.

The principal measured contributions are the workspace characterisation, with its negative result and its quantified mechanical remedy; the account of what degraded sensing costs and which two channels matter most; the finding that the deployed clearance margin and the planner’s collision test disagree across the entire working band, resolvable laterally but not fore-and-aft; and the observation that two independent constraints — clearance and wrist feasibility — place the usable boundary in the same place.

The methodological contribution is the discipline of validating an instrument against a known answer before reporting a surprising measurement taken with it. This arose from necessity: across the project, unexpected results were more often defects in measurement than in the system, and requiring every analysis routine to recover a constructed answer caught a substantial number of errors that would otherwise have entered the record.

7.1 Future work

The immediate steps are set by the limitations rather than by ambition, and are ordered by how much each would improve the system.

Repair the two identified master channels. Regression shows the left arm retains no reach observable without them, and every direct-teleoperation baseline depends on them. This is the single highest-value action.

Measure control-path latency. The instrumentation is small — seven timestamps and a sequence number already partly present — and it is the only one of the planning report’s targets that is both untouched and immediately measurable. The result must be reported as a lower bound.

Apply the mount change and re-derive. The sweep quantifies the gain; applying it invalidates the home pose, the workspace anchor and every clearance figure, so it should be one deliberate pass with full re-verification.

Ingest and document the master hardware record. The firmware, CAD and bill of materials exist outside the repository; committing them would supply the pin map, ADC resolution and fabrication parameters that [Chapter 3](#) currently cannot state.

Then run the study. The design is complete and costed at sixteen dyads over a 167 min session, comparing direct teleoperation against shared autonomy across all five tasks.

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Chapter A

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Chapter B

Verified Task Scenarios

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Chapter C

Verification Recordings

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